Dante Lok (Lok Fu On)

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RESEARCH INTERESTS

Mathematically structured learning, and how training interventions change what a network can compute: function-preserving model growth, continual learning, geometric and equivariant architectures, and visual and abstract reasoning. I pair theoretical analysis with large-scale experiments.

PUBLICATIONS AND PREPRINTS

What Does Model Growth Really Add? Functional Capacity Beyond Parameter Count
Dante Lok. *Preprint, 2026 (under review)* arXiv:2607.14571 | code

RESEARCH EXPERIENCE

Senior AI Engineer, Votee AI Sep 2025 – Oct 2026

- Defined and lead the research team's scientific agenda: how training interventions (model growth, fine-tuning, continual learning, RL) change a network's functional capacity, targeting reasoning with mathematical structure across math, science, and vision.
- Authored *What Does Model Growth Really Add?* Trained Transformers from scratch for model-growth experiments (253M → 857M), built a matrix-free functional-Jacobian rank estimator, and proved rank additivity under transversality; all 36 added directions survived continual learning despite catastrophic forgetting.
- Leading ongoing research on physical reasoning in vision, currently gravity-aware and equivariant object dynamics.
- Post-trained a 35B Mixture-of-Experts LLM (CPT → SFT → DPO → GRPO) for Cantonese and Hong Kong cultural understanding, improving HKCanto-Eval phonological reasoning from 27% to 84% without degrading English benchmark performance.
- Trained a Cantonese flow-matching TTS model from scratch on 3,400 curated hours filtered from a 21k-hour corpus, with a polyphone- and tone-sandhi-aware Jyutping G2P front-end.
- Built a production multi-agent orchestration platform and sandbox runtime for isolated code and tool execution.
- Mentored 2 research and engineering interns from project scoping through delivery.

AI Engineering Lead, Omnichat Oct 2024 – Aug 2025

- Led four engineers and two product managers in defining and executing the AI product roadmap.
- Architected multi-agent systems integrating retrieval, multimodal understanding, workflow automation, behavioral analytics, and sentiment analysis; deployed in client WhatsApp chatbots with an estimated thousands of monthly users.
- Developed end-to-end voice-agent and generative-media prototypes, covering model integration, evaluation, and production delivery.

AI Engineer, LifeHikes Feb 2020 – May 2020; Mar 2021 – Jan 2024

- Trained speech, text, and video models in TensorFlow and PyTorch (voice-modulation classifiers, BERT-family text classifiers, MediaPipe with YOLO/R-CNN video analysis), automating all 12 targeted analysis tasks.
- Cut video and voice analysis latency from 20+ s to 1–3 s for the mobile app: compiled TensorRT engines from ONNX exports for GPU serving, and served smaller text classifiers on CPU with ONNX Runtime to free limited GPU capacity.
- Implemented and productionized research methods from HKUST, working with their faculty and PhD researchers: voice modulation (VoiceCoach, CHI 2020), multimodal emotion coherence (EmoCo, IEEE VIS 2019), and engagement recognition (CMOSE, CVPR Workshops 2024), including methods covered by 2 granted US patents (US 11,282,297; US 11,282,508).

Analyst Programmer, Machine Learning, ICONTEK Technology Jun 2018 – Nov 2019

- Machine translation research with Prof. Alex Waibel (Carnegie Mellon University).
- NLP and speech analysis research on English and Chinese with Seq2Seq and attention models.

Data Analyst, Citrix Jan 2018 – Jun 2018

- Initiated the first machine learning project for sales prediction and lead identification.

EDUCATION

The Hong Kong Polytechnic University, BSc in Statistics and Computing 2017

SKILLS

Research and ML: PyTorch, JAX, Transformers, continual learning, geometric deep learning, speech and language modeling, multimodal learning
Programming: Python, C++, SQL, JavaScript
Systems: CUDA, Triton (GPU kernels), Triton Inference Server, vLLM, TensorRT, ONNX, Docker, Kubernetes, AWS